



Databases

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Lecture Outline

- Lecture I** – past, present, future
- Lecture II – schema design, SQL
- Lecture III – scalability & reliability challenges



The challenge

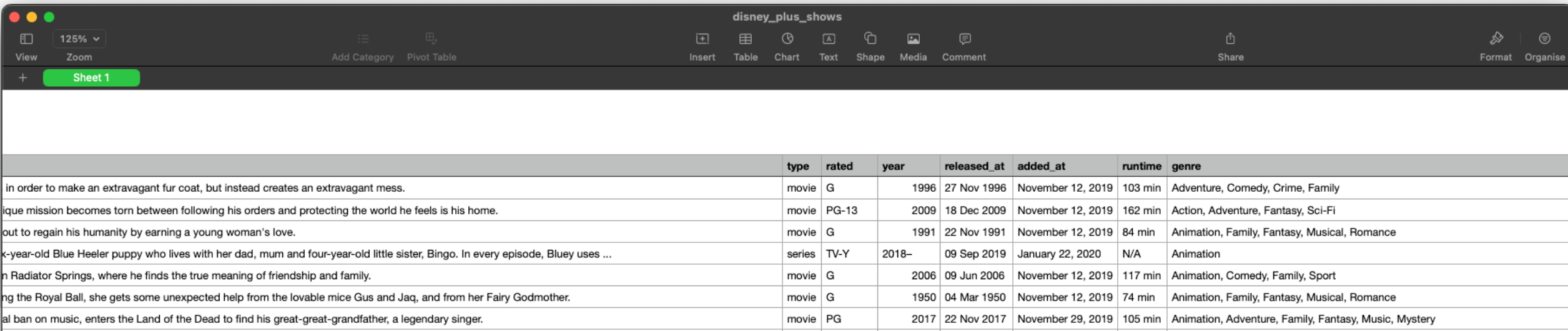
How do you store information (data)?

The past – paper



The access time is bad,
The cost is high,
The data is *not digital*,
Concurrent access is limited,
etc.

The past – flat files



The screenshot shows a spreadsheet application window titled "disney_plus_shows". The interface includes a top menu bar with options like View, Zoom, Add Category, Pivot Table, Insert, Table, Chart, Text, Shape, Media, Comment, Share, Format, and Organise. Below the menu is a toolbar with icons for these functions. The main area displays a table with the following data:

	type	rated	year	released_at	added_at	runtime	genre
in order to make an extravagant fur coat, but instead creates an extravagant mess.	movie	G	1996	27 Nov 1996	November 12, 2019	103 min	Adventure, Comedy, Crime, Family
ique mission becomes torn between following his orders and protecting the world he feels is his home.	movie	PG-13	2009	18 Dec 2009	November 12, 2019	162 min	Action, Adventure, Fantasy, Sci-Fi
out to regain his humanity by earning a young woman's love.	movie	G	1991	22 Nov 1991	November 12, 2019	84 min	Animation, Family, Fantasy, Musical, Romance
x-year-old Blue Heeler puppy who lives with her dad, mum and four-year-old little sister, Bingo. In every episode, Bluey uses ...	series	TV-Y	2018–	09 Sep 2019	January 22, 2020	N/A	Animation
n Radiator Springs, where he finds the true meaning of friendship and family.	movie	G	2006	09 Jun 2006	November 12, 2019	117 min	Animation, Comedy, Family, Sport
ng the Royal Ball, she gets some unexpected help from the lovable mice Gus and Jaq, and from her Fairy Godmother.	movie	G	1950	04 Mar 1950	November 12, 2019	74 min	Animation, Family, Fantasy, Musical, Romance
al ban on music, enters the Land of the Dead to find his great-great-grandfather, a legendary singer.	movie	PG	2017	22 Nov 2017	November 29, 2019	105 min	Animation, Adventure, Family, Fantasy, Music, Mystery

The access time is better,

The cost is lower,

The data is digital,

Concurrent access is limited,

The data quality is... varied (see the date formats)

The file structure is... inconsistent

Newlines can be \n or \r\n

Files can have different charsets – Windows, ISO, UTF without BOM, UTF with BOM

The entire global
financial system

Excel



Why Excel is not a database?

- No data integrity enforcement
- No transaction support
- Poor concurrency
- Limited to 10^6 rows
- No query optimization
- No access control
- No backups, etc

Me: didn't save for 30 minutes
Excel: stops responding
Me:

Excel 97-2003
files still working



Excel is not a database
Excel is not a database
Excel is not a database



The avalanche of data ...

Humans are generating lots and lots of data

We need to have an organised way to collect and analyse this data

There are many concurrent users for many datasets

We can do better than Excel

DATABASES!



Database

A **database** is:

- an organized collection of data or a type of data store based on the use of a **database management system (DBMS)**,
- the software that interacts with end users, applications, and the database itself to capture and analyze the data.
- The DBMS additionally encompasses the core facilities provided to administer the database. The sum total of the database, the DBMS and the associated applications can be referred to as a **database system**.

Most often, when people mention a database, they mean the dataset + the software

Database Types

Relational Databases (RDBMS)

NoSQL (Key-Value, Document) – not only SQL

Graph, Hierarchical – data is stored as nodes and edges

In-Memory – ultra-fast lookups

Text-Search – Full Text indexes

Spatial – GIS systems, maps, etc.

Ledger / blockchain – bookkeeping, finance, etc.

Vector – AI workloads

Columnar / Time-Series – analytical workloads

NewSQL – a combination of traditional approach with NoSQL flexibility

There is no one-size fits all solution. Choose your databases wisely!

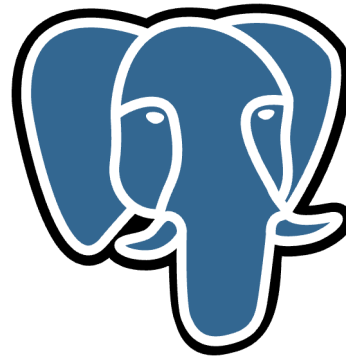
The present – relational databases

You probably heard about some of the following:

Oracle, PostgreSQL, MySQL (MariaDB), Microsoft SQL Server, SQLite

Each implementation differs, has other design choices, but the general idea stays the same.

We'll focus on **PostgreSQL** and we'll use it during the exercise session



The present – relational databases – cont.

A relational model:

- **organizes data into one or more tables (or "relations") of columns and rows**
- a unique key identifying each row

Generally, each table/relation represents one "entity type" (such as customer or product). The rows represent instances of that type of entity (such as "Lee" or "chair") and the columns represent values attributed to that instance (such as address or price).

Student ID	Name	Residence
123	Alice	FR
456	Bob	UK
789	Carol	AA

The present – relational databases – cont.

We must set data types (e.g. ID is a number, date of birth is a date, name is a text)

We can set constraints (e.g. name cannot be null,
education level could be one of *{PhD, Master, Bachelor, null}*)

We can set foreign keys (e.g. country of residence is one of the entries in countries table)

We can set unique keys (e.g. ID has to be unique)

<i>unique number</i> Student ID	<i>not null text</i> Name	<i>char(2)</i> Residence
123	Alice	FR
456	Bob	UK
789	Carol	AA

CountryID	Name
BR	Brazil
FR	France
CL	Chile

The present – relational databases – cont.

We can set **primary keys** (e.g. ID)

- Primary Key has to have unique values in the table
- Primary Key cannot be null
- Primary Key will be indexed to ensure fast access
- Primary Key can span multiple columns $\Rightarrow$ composite key

Primary Key
number

Student ID	Name	Residence
123	Alice	FR
456	Bob	UK
789	Carol	AA

not null
text

char(2)

Primary Key

CountryID	Name
BR	Brazil
FR	France
CL	Chile

Postgres data types

Category	PostgreSQL data types
Numeric Types	smallint, integer, bigint, decimal, numeric, real, double precision
Character Types	char(n), varchar(n), text
Date/Time Types	date, time, time with time zone, timestamp, timestamp with time zone, interval
Boolean Type	boolean
JSON & XML Types	json, jsonb, xml
UUID	uuid

**Each column is allowed to contain NULL unless you specify NOT NULL constraint
(or the column is a part of primary key)**

How many states can a boolean have?

The boolean type can have several states: “true”, “false”, and a third state, “unknown”, which is represented by the SQL null value.

TRUE AND NULL evaluates to... NULL

NOT NULL evaluates to... NULL

Be aware of that when you run comparisons.

You can use “IS NULL” operator to check if the column’s value is NULL.

You can force “TRUE/FALSE” behaviour by using a “NOT NULL” constraint.

How is data stored on disk?

Student ID	Name	Residence
123	Alice	FR
456	Bob	UK
789	Caroline Anne	US

Data storage is done in a row-based way, in 8 KB blocks called pages

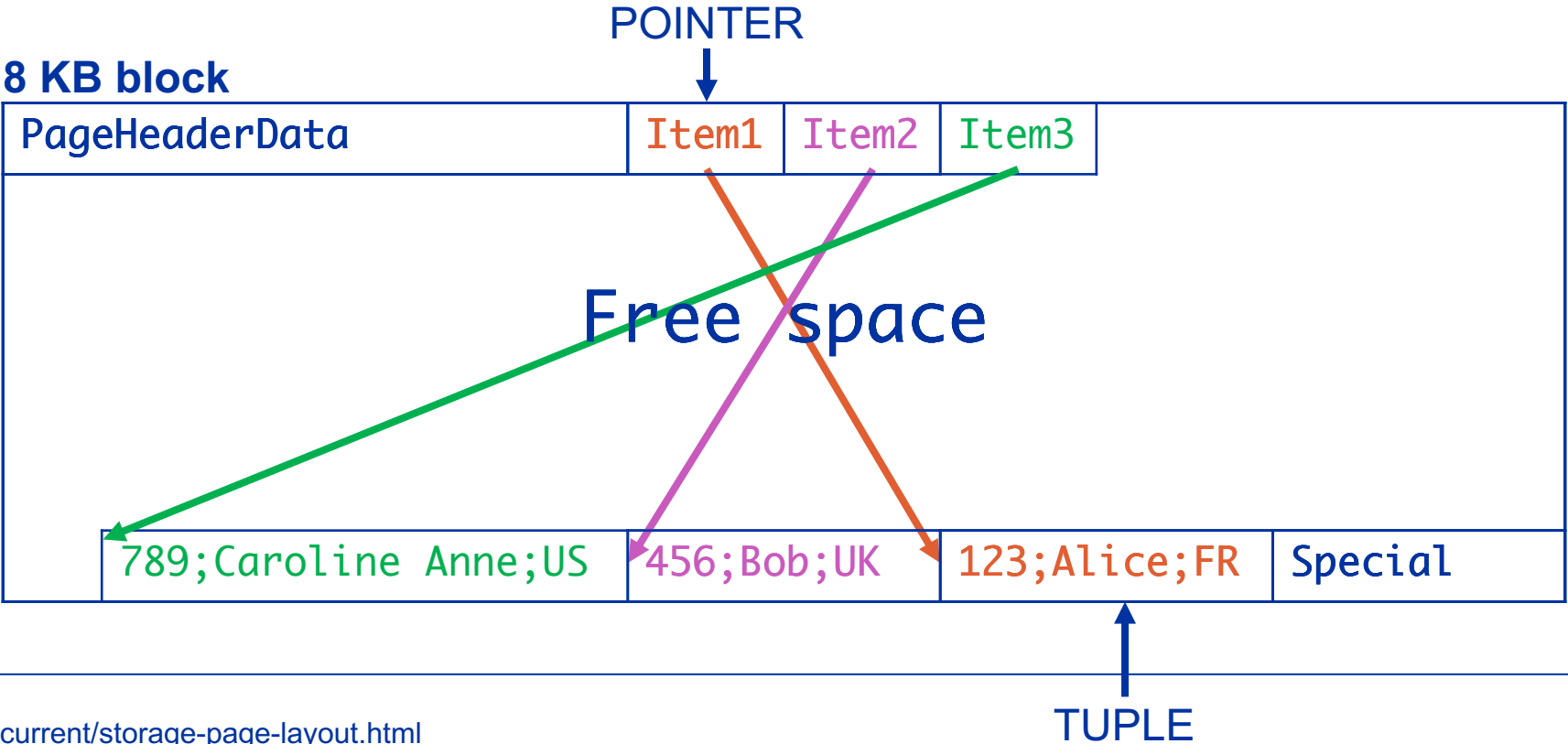
Alice Bob Caroline Anne char(16)

AliceBobCarol varchar(16)

PageHeaderData contains some internal information like:

- location of the last pointer
- location of the "first" tuple

We can calculate free space

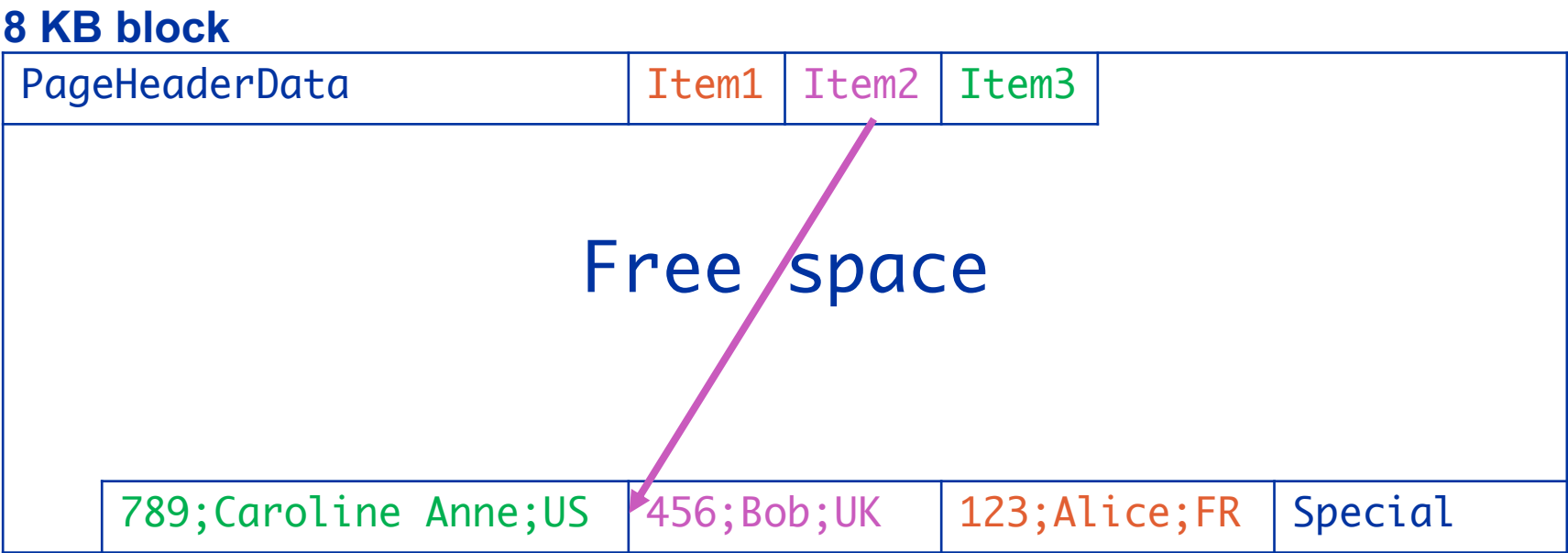


What if you update your data?

Let's update Bob to Bob Junior

Student ID	Name	Residence
123	Alice	FR
456	Bob Junior	UK
789	Caroline Anne	US

1. We need to read the old row contents



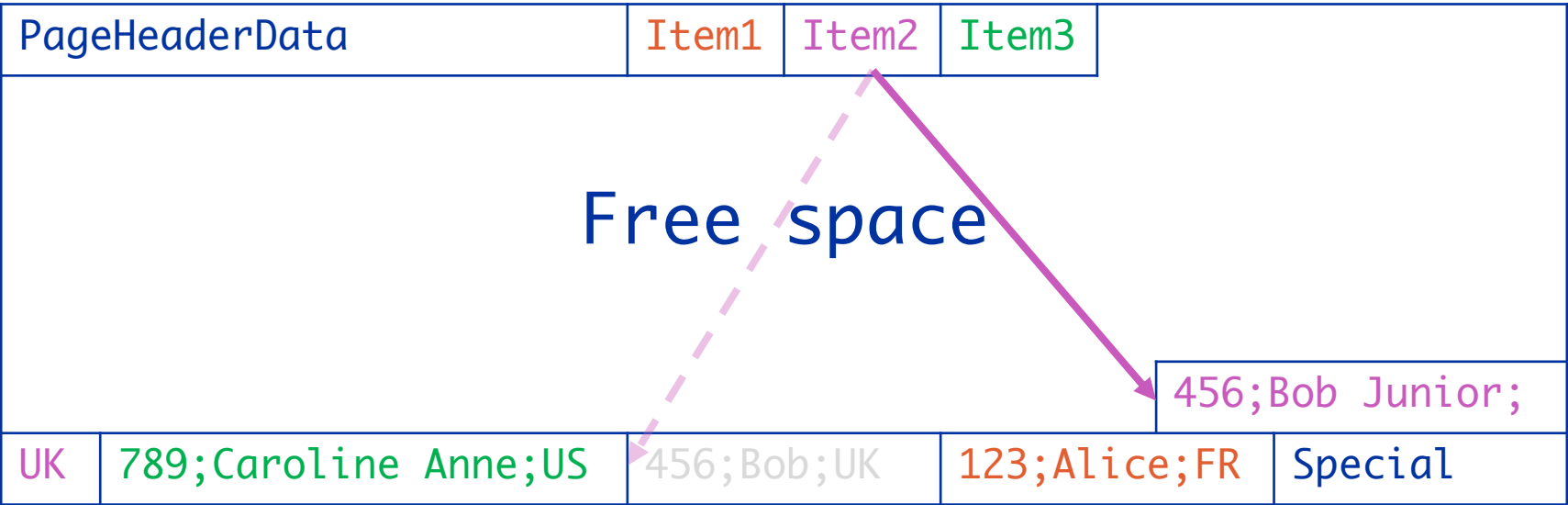
What if you update your data?

Let's update **Bob** to **Bob Junior**

Student ID	Name	Residence
123	Alice	FR
456	Bob Junior	UK
789	Caroline Anne	US

- 1. We need to read the old row contents
- 2. We add new version of the row
- 3. We change the pointer, thus invalidating old content
- 4. Old content will be cleaned up later

8 KB block



How do you locate data in huge datasets?

Stage—Standard

Stage Studios ILGWU 106W39 BR yant 9-1163
Stage W Barden Mrs. photogr 780MadAv RE gent 4-7542
Stage Wm S K 270W11 CH elsea 3-3875
Stagecraft Studios scenery 1828AinStrdmAv ED gecomb 4-2224
Stagecrafters Assn Inc 1650Bway CO lumbs 5-8808
Stagecrafters Dinner Club 115W49 CI rcle 5-9485
Stager A 401WEndAv SC huyir 4-6013
Stager Emmanuel 201E87 AT watr 9-8017
Stager Lillian M 525 1Av LE kingtn 2-8816
Stager Stanley R 280RivDr AC admy 4-2623
Stager Stanley R ynts 207 4Av GR amrcy 5-1355
Stagg A M Lumbr Co Inc 143 OceanAv JCNJ (Dial the Operator) Delaware 3-7220
Stagg Co 259W14 WA tkns 9-8580
Stagg D 247E39 MU rryhil 4-5851
Stagg De Peyster 32BeekmPl PL aza 3-6567
Stagg Erna Mrs 610W139 AU dubn 3-1117
Stagg Fredk L Mrs interiors 38E57 PL aza 3-1300
Stagg Geo T Co 350 5Av CH ickng 4-7200
Stagg Jas Hardwick CPA 141Bway BA rcl 7-5580
Stagg Mather&Hough acctnts 141Bway BA rcl 7-5580
Stagg Max coats 135W26 LA ckwna 4-1864
Stagg Mildred Mrs 1060PkAv SA crmnto 2-8561
Stagg Plumbing Supl Co Inc 102Geary Bklyn EV ergn 8-1204
Stagg Rosalie E Mrs 245E21 GR amrcy 5-1293
Staggers Mabel Stuart 235E22 GR amrcy 7-2196
Staggs Aluminum Limb Co 1931Bway TR afalgr 7-9156
Stag's Head Tavern 880AmstAv AC admy 2-9755
Stagshhead Cafe 3476Bway AU dubn 3-9502
Stahel Edw P&Co com merchants 354 4Av CA ledonia 5-5798
Stahel Rose Mrs 825WEndAv AC admy 4-5718
Staheli Rietmann&Co 95MadAv AS hind 4-8512
Stahl Adolf trip bags 50UnSq ST uyvnt 9-5330
Stahl Anna M Mrs 504W146 ED gecomb 4-2958
Stahl Anne Mrs 661W180 WA dwth 3-3424
Stahl Benj 735W172 WA dwth 3-7505
Stahl C W 495W186 WA dwth 3-4611
Stahl Carl DDS b 1697Bway CO lumbs 5-5741
Stahl Chas 198NapAv LO rrain 7-8788
Stahl&Co sewing machs 756 6Av WA tkns 9-0650
Stahl D Mrs 521W189 WA dwth 3-0503

Shop By Telephone—The Classified Directory Tells "Where To Buy It"

Stahr Paul 510W183 WA dwth 3-2894
Stahrenberg Peter H prntng 147E116 LE high 4-1833
Stalano F fruit 480 6Av GR amrcy 5-8737
Stalano Ralph vgs 1545 1Av RE gent 4-6922
Stalib Carl 352CPkV RI versde 9-5596
Stalib Chas b 377Bway CA nal 6-0077
Stalibrite Letter Co 42W30 BO gardus 4-0639
Staiger Chas G jewls 655 5Av PL aza 3-6705
Staiger Engraving Co 52Beekm BE ekmn 3-5346
Staiger Hugo M jwlr 527 5Av MU rryhil 2-6378
Staiger&Max Inc jewls 655 5Av PL aza 3-6705
Staiger&Sons jwlr 527 5Av MU rryhil 2-6378
Staiger Wm E jwlr 527 5Av MU rryhil 2-6378
Stalios Bros grocs 356W40 BR yant 9-9382
Stalman Frieda 460W24 CH elsea 3-0932
Stalman Martin G 460W24 CH elsea 3-0932
Stain-Ox Co Inc 215Mntagu Bklyn MA in 4-0466
Stainer Z 120E2 GR amrcy 3-2333
Staines A Mrs 251W95 RI versde 9-5267
Stainless Bar&Restnt Equip Co Inc 147Bway CA nal 6-8342
Stainless Electro Plating Co 67 Sprng CA nal 6-3955
Stainless Metals Inc 31-49 12th LongIslandCity RA venswd 8-7555
Stainless Razor Blade Corp 475 5Av LE kingtn 2-0521
Stainless Steel Weatherstrip Co 108W18 WA tkns 9-3318
Stainpruf Distributg Corp 342MadAv MU rryhil 2-0274
Stainton A E 819W180 WA shHts 7-3618
Stainton Estelle Mrs 132W74 TR afalgr 7-1436
Stair Alastair A anties 59E57 VO lunter 5-7620
Stair&Andrew Inc See Stair&Co Inc
Stair&Co Inc antq furn 59E57 VO lunter 5-7620
Stair Covering Specialty Co 418W125 MO numnt 2-3417
Stair David hndbgs 12W32 LO ngarc 5-3325
Stair Edw b 31Nasau BA rcl 7-0976
Stair Edw 22E36 MU rryhil 5-0942
Stair G W b 4E46 MU rryhil 2-1149
Stair Geo W books 7W49 CI rcle 5-8040
Stair-Goldberg advtg 136Libty RE ctr 2-9713
Stair Nancy Mrs 150CentPkS CI rcle 7-4882
Staite Geo 505W162 WA dwth 8-0224

Stamer Edw butcher 492LenxAv AU dubn 3-4369
Stamer Edw butcher 2714 8Av ED gecomb 4-9030
Stamer Eric H 75PostAv LO rrain 7-8145
Stames Bros 593ColAv SC huyir 4-4480
Stames Gus grocr 593ColAv SC huyir 4-4480
Stametal Inc 101MaidenLa WH itchl 4-3229
Stamford Bakeries Inc 2124 8Av MO numnt 2-7116
Stamillman Louis M 250FtWashAv WA shHts 7-1791
Stamis Press 395Pri BE ekmn 3-3275
Stamler Bros&Jacobson Inc clks sts 266W37 CH ickng 4-2072
Stamler Florence R Mrs 760WEndAv AC admy 4-1829
Stamler I dntst 2700BrnxPkE OL invl 5-7887
Residence 851E181 RA ymd 9-1455
Stamler I furrier 18W56 CI rcle 7-6769
Stamler Saml b 419E54 PL aza 3-9876
Stamm A L&Co brkrs 120Bway RE ctr 2-6800
Stamm Alfred L brkr 120Bway RE ctr 2-6800
Stamm Alfred L 300CentPkW SC huyir 4-6963
Stamm Chas G HotelLucerne EN dicot 2-7100
Stamm Ella Mrs 519W189 WA shHts 7-7655
Stamm Emilie 17E9 GR amrcy 3-0125
Stamm Everard M C 23W73 EN dicot 2-3500
Stamm G Edw b 140Nasau BE ekmn 3-5522
Stamm Geo L lawyer 140Nassau BE ekmn 3-5522
Stamm Henri b 99Hudson WA lkr 5-2387
Stamm J D 35SutnPl So EL dorado 5-6898
Stamm Jack 131RenettAv WA dwth 8-1826
Stamm Jules 10RiDrW WA dwth 3-0454
Stamm L Elkin Mrs 19DeegnPl LO rrain 7-5046
Stamm Lillian Mrs 262CPkV SC huyir 4-1977
Stamm Louis S b 1372Bway WI sconth 7-7800
Stamm Paul 550PkAv RE gent 4-0627
Stamm Rudolph engrvr 64Nasau BO wGrn 9-2795
Stamm Rudolph 301E21 ST uyvnt 9-3217
Stamm Wm H 160W73 TR afalgr 7-6700
Stammelman Mortimer J Dr b 119W23 CH elsea 2-0058
Stammer E Mrs 216E12 GR amrcy 5-1091
Stammer P bk store 61 4Av ST uyvnt 9-2448
Stammer Saml dntst str 125Cent WA lkr 5-3993
Stammers J C Dr 330W72 TR afalgr 7-6526
Stammiller Wilber lwr 170Bway CO rting 7-3434
Stamoolis&Co Inc food prods 230E30 LE kingtn 2-3762
DI qby 4-7890
WA ndrblt 6-1092
BE ekmn 3-4076

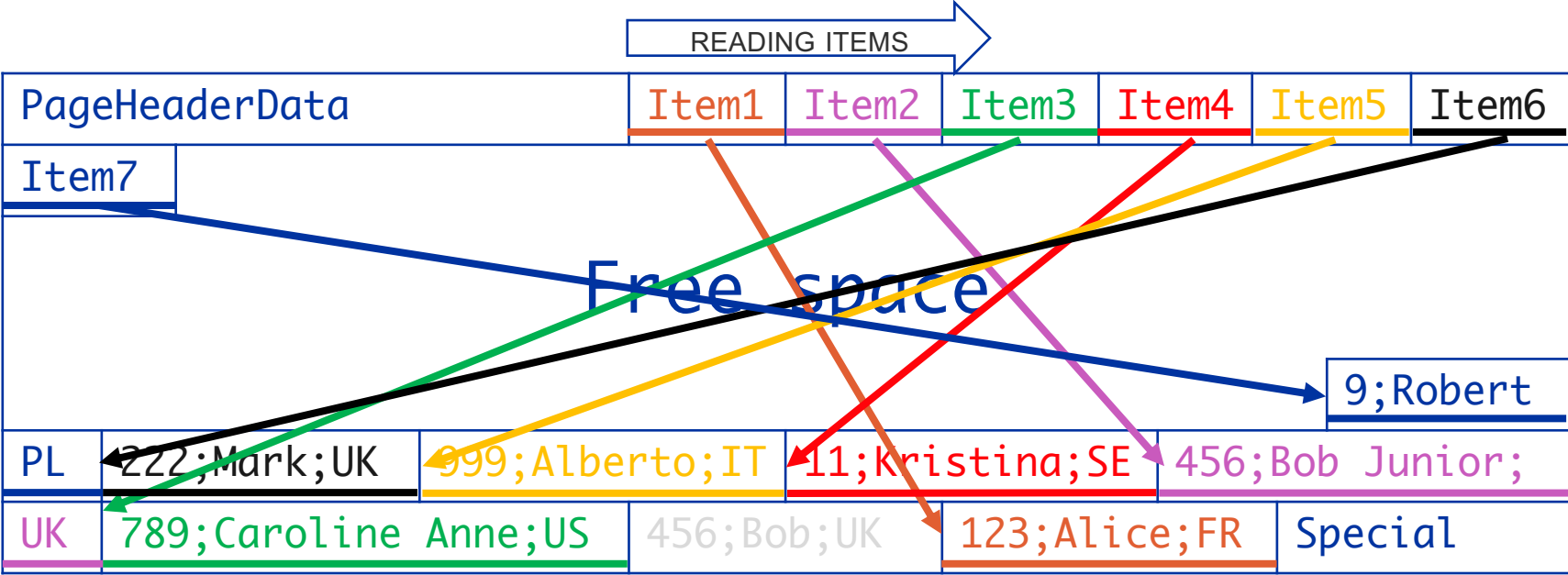
"b" following name denotes business service

970

Standard Appraisal Co 6Church CO rting 7-4493
Standard Arch Co fireprfg 245W18 CH elsea 3-5176
Standard Arm Co awning supls 152Blecker GR amrcy 7-0456
Standard Arm Wks 79Craby CA nal 6-6579
Standard Aromatics 1WForestAv EnglewoodNJ (Dial the Operator) Englwd 3-5770
Standard Artists Bur 145W45 LO ngarc 5-7721
Standard Asbestos Co Inc 69Beekm BE ekmn 3-4567
Standard Auto Radiatr Co 516W48 CI rcle 6-7698
Standard Auto Repr&Parts Svce Co 5-8446
Standard Auto Spring Co 9Av&207 LO rrain 7-9423
Standard Bag&Burlap Co 24Bway Bklyn EV ergn 7-1869
Standard Bag Corp 295MadAv CA ledonia 5-8388
Standard Bag Corp 295MadAv LE kingtn 2-6138
Standard Bakers Cooperative Inc 23W56 CI rcle 5-8044
Standard Bank of South Africa Ltd 67Wall BO wGrn 9-1464
Standard Beauty Parlor Supl Co 20W33 CH ickng 4-6367
Standard Beauty Supl Co 20W33 CH ickng 4-6367
Standard Beef Co 2267 7Av AU dubn 3-8290
Standard Beer Pump Svce 1555MacombRd TR emnt 2-4433
Standard Beveling&Silvering Co Inc 4137 3Av TR emnt 2-6280
Standard Bias Binding Co 118W22 WA tkns 9-0514
Standard Bien Button Co Inc 146W29 WI sconth 7-6806
Standard Bitulithic Co 50Chrch CO rting 7-1267
Standard Bleachery&Prntng Co 40Worth WO rth 2-0136
Standard Bloomer Co 38W32 CH ickng 4-8191
Standard Blouse Inc 142W36 CH ickng 4-1137
Standard Body Corp 430E104 LE high 4-1260
Standard Boiler Svce 161W4 CH elsea 3-0508
Standard Book Co 240MadAv CA ledonia 5-8999
Standard Book Co pubes 381 4Av MU rryhil 5-0216
Standard Bkbnng Co 27W24 CH elsea 2-7017
Standard Bookinn Office 100W57 CI rcle 7-3206
Standard Brake Svce Corp 102051NchAv WA dwth 3-9284
Standard Brake Svce Inc 229 10Av CH elsea 2-9191
Standard Brake Shoe&Foundry Co 30Church CO rting 7-7633
Standard Brake Sta Inc 614 1Av MU rryhil 4-9163

Imagine you want to access row ID=11

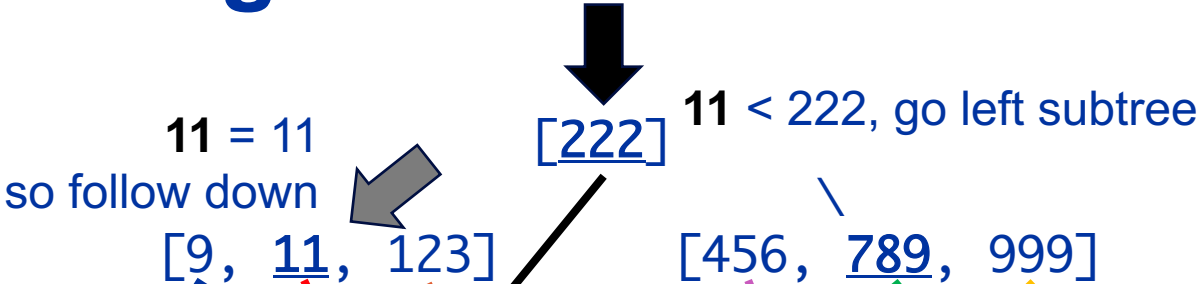
If we wanted to get one row only, we would stop at Item4.
But if we didn't – we need to continue



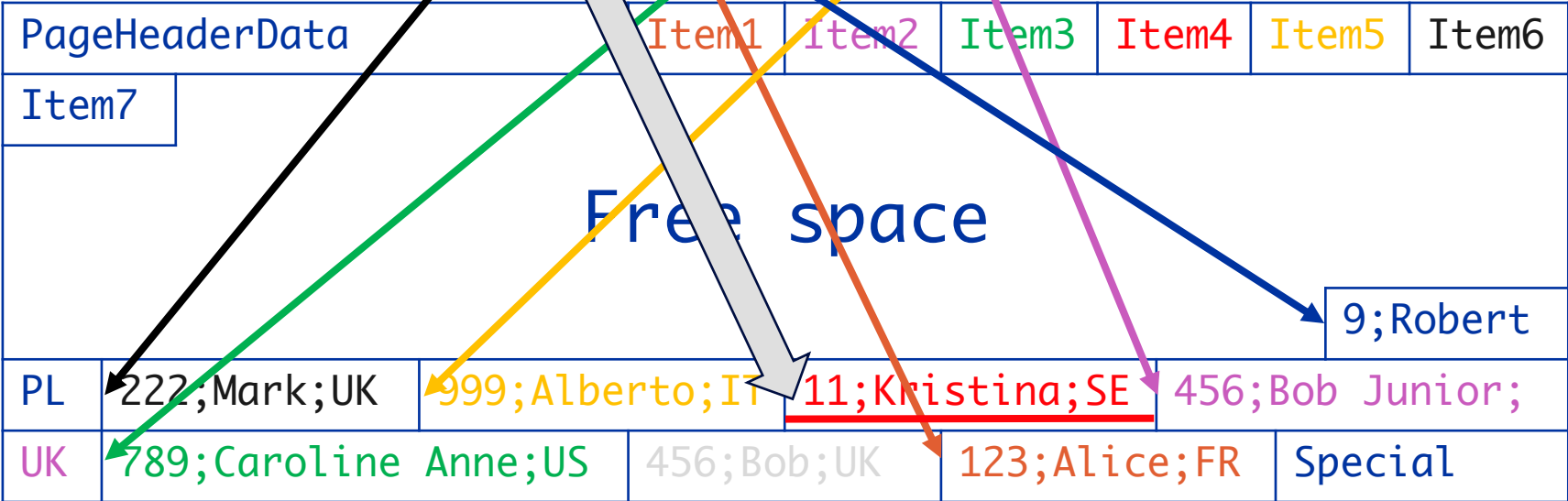
That's a Sequential Table Scan operation, also known as Full Table Scan.
Slow, but always possible.

Let's try implementing a B-tree Index on ID column

9	Robert	PL
222	Mark	UK
999	Alberto	IT
11	Kristina	SE
456	Bob Junior	UK
789	Caroline Anne	US
123	Alice	FR



You want to access row ID=11 now.
That's an Index Scan operation



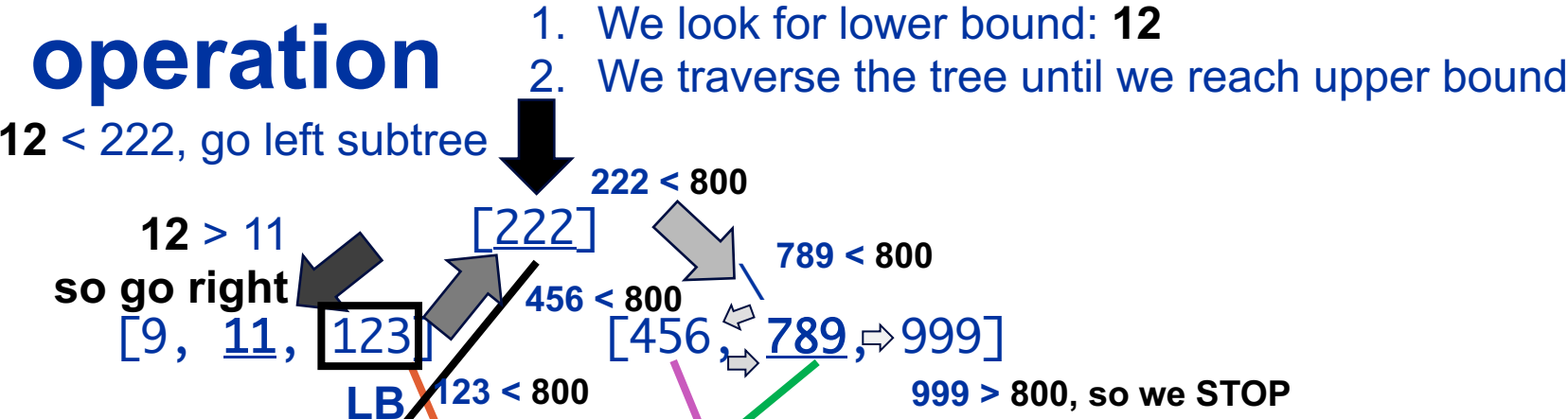
Index range scan operation

9	Robert	PL
222	Mark	UK
999	Alberto	IT
11	Kristina	SE
456	Bob Junior	UK
789	Caroline Anne	US
123	Alice	FR

You want to access rows with
ID BETWEEN 12 AND 800

Result set:

123	Alice	FR
222	Mark	UK
456	Bob Junior	UK
789	Caroline Anne	US



PageHeaderData	Item1	Item2	Item3	Item4	Item5	Item6
Item7						
						9;Robert
PL	222;Mark;UK	999;Alberto;IT	11;Kristina;SE	456;Bob Junior;		
UK	789;Caroline Anne;US	456;Bob;UK	123;Alice;FR	Special		

Indexes

Indexes speed up data access by avoiding full table scans: $O(\log n)$ vs $O(n)$

The results are already sorted

Indexes take up space

Indexes cause an overhead when data is changed (on each operation)

Index scans are random IO operations, sometimes sequential access to a table can be faster. Sometimes indexes won't be used (e.g. low selectivity or large result sets).

Planner / Optimizer

There is more than one way to access information, each providing the same & correct results. Some decisions that need to be taken: which index to use (if any), join strategies, join order, etc.

For complex queries there could be hundreds of possibilities.

RDBMSes use different techniques & tricks to find the optimal “execution plan”. In the past – these were rule-based systems (*“prefer index over table scan if index size is smaller than table size”*).

The RDBMSes are collecting statistics to better “understand” the data and predict the results

Most RDBMS now implement a **Cost-Based Optimizer**: this execution plan will use x amount of CPU, y amount of I/O and z amount of memory. Total cost is calculated based on those. The cost value doesn't have a unit, it's value doesn't mean much in real life – it's only to be used for comparison with other cost estimations.

It's probable that a query with cost of 1000 could run for 10 seconds and a query with cost 1500 will run for 5 seconds, but for sure both of them will be faster than a query with a 100k cost.

We need a way to “communicate” with the database

SQL – a language used to manage data, especially in a relational database management system (RDBMS)

SQL will let us:

- create / modify / drop tables
- insert / update / delete rows
- grant permissions
- query our data

The NoSQL databases might use different interfaces (REST api, custom protocols, etc.)

Database Transactions

Bob wants to send 100 USD to Alice. This consists of two operations:

- Increase Alice's account balance by 100 USD
- Decrease Bob's account balance by 100 USD

What if the software / database / server crashes between those operations?

We can group operations into transactions. Each transaction can finish with a:

COMMIT; – data is saved durably and will survive eventual crashes

ROLLBACK; – the changes from transaction are “cancelled” as if the operation never happened

Rollback is the default behaviour if your application crashes or there is a network connectivity error.

ACID properties

ATOMICITY – each transaction either occurs in full or not at all

CONSISTENCY – each transaction changes the database from one consistent state to another; at any point in time all the data in the database will follow all the rules of constraints, foreign keys, uniqueness, etc.

ISOLATION – sessions can only see data committed by other transactions. Only one transaction can modify a row at a time. Sometimes your data will be locked. DB systems offer many isolation levels to offer flexibility (read uncommitted, read committed, repeatable reads, serializable).

DURABILITY – a committed transaction will survive permanently, even in case of failures; most frequently RDBMS will return „COMMIT OK” message only if the data is persisted to disk

Why do we keep multiple versions of the rows?

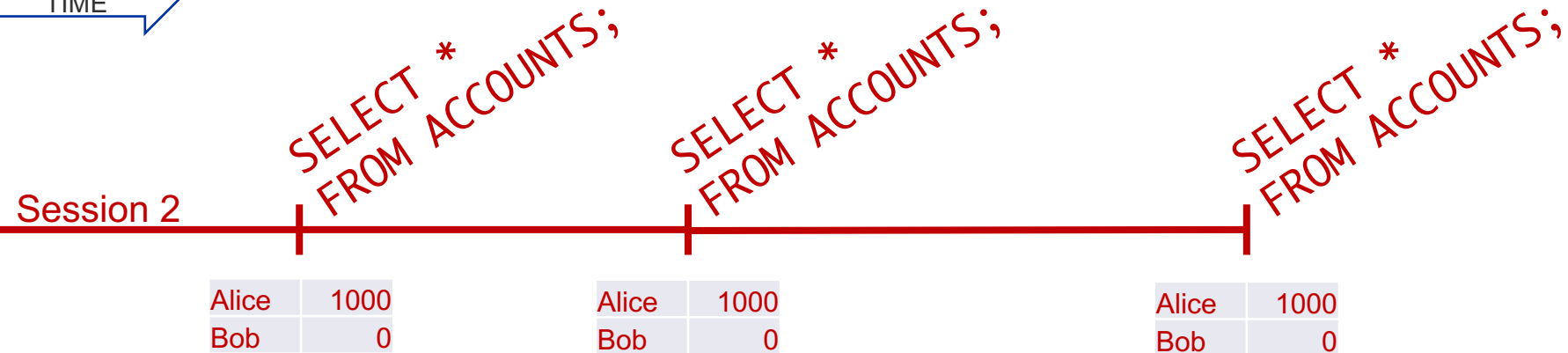
Alice	1000
Bob	0



Alice	2000
Bob	0

Alice	2000
Bob	0

After rollback the version of the row with value 2000 can be removed



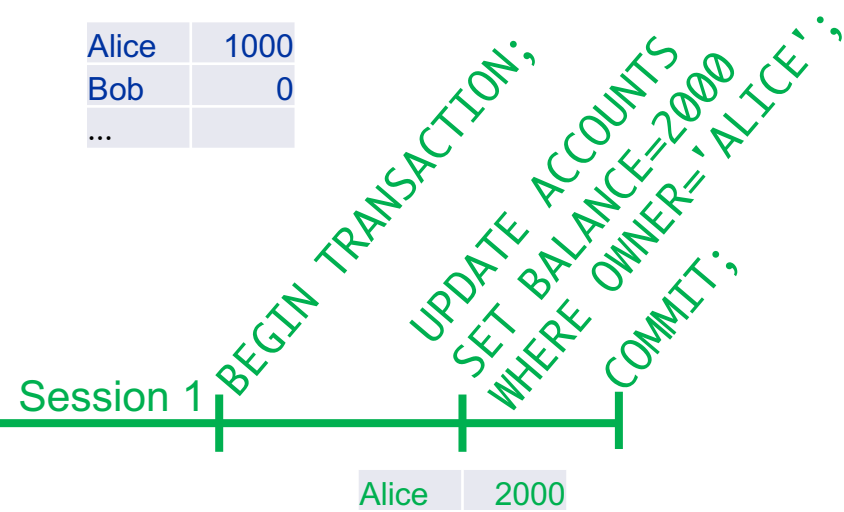
Alice	1000
Bob	0

Alice	1000
Bob	0

Alice	1000
Bob	0

MVCC – consistency and isolation

Alice	1000
Bob	0
...	



SELECT *
FROM ACCOUNTS
ORDER BY NAME DESC;

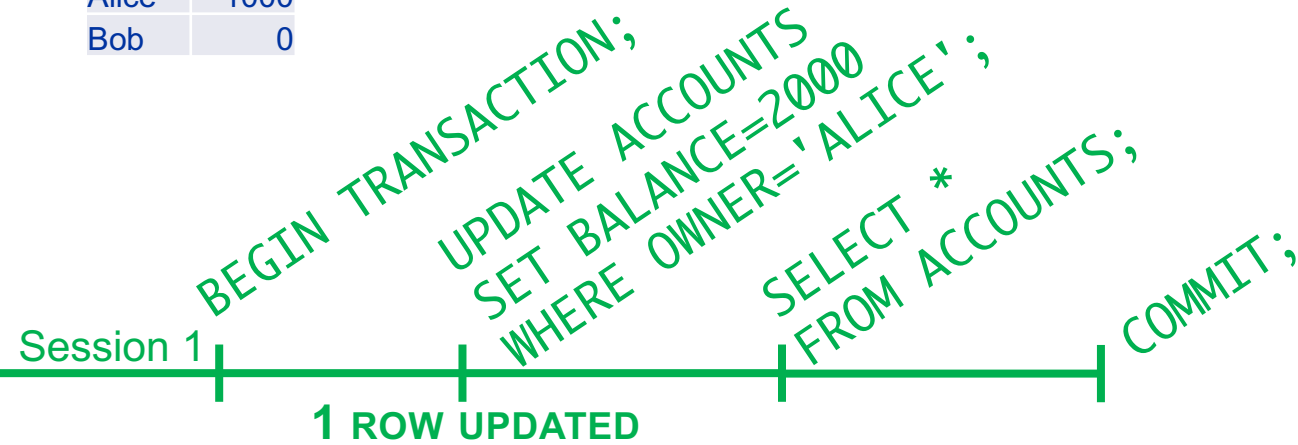


Rose	999
Patricia	111
Luke	666
Helena	123
Federico	333
Cedric	21
Bob	0
Alice	1000



MVCC – row locks

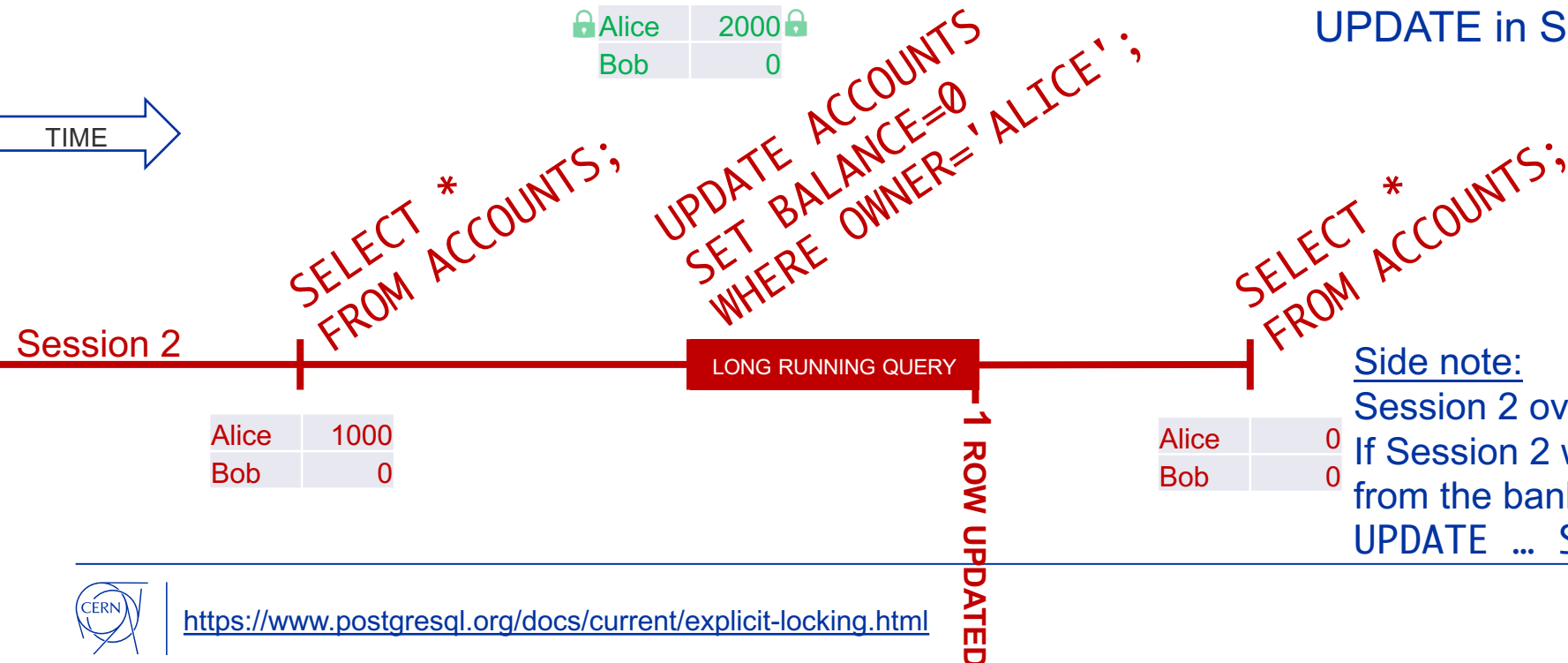
Alice	1000
Bob	0



In most RDBMSes:

- only one operation can update a row at any time
- multiple operations can read a row
- RDBMS keeps track of different type of locks (exclusive, shared) and it's interoperability

COMMIT finishes the transaction
The locks are released
UPDATE in Session 2 can finish



Side note:

Session 2 overwrote Session 1's increase of balance
If Session 2 was the operation of spending 1000 \$
from the bank account, it should be written as:
UPDATE ... SET BALANCE=BALANCE-1000

The future

NoSQL (Key-Value, Document) – not only SQL

Vector – AI workloads

Columnar / Time-Series – analytical workloads

NewSQL – a combination of traditional approach with NoSQL flexibility



NoSQL databases

- **Schema-flexible or schema-less**
- **Designed to scale across many nodes**
- **BASE consistency (Basically Available, Soft state, Eventually consistent) instead of ACID**
- **They could be using other access methods than SQL**
- **Horizontal scaling, some offer interesting synchronization (e.g. for mobile/offline use)**

Examples:

- **Document Stores – MongoDB, Couchbase, CouchDB**
- **Key-Value – Redis, DynamoDB**
- **Graph – Neo4j**

NewSQL databases

- **Defined schema**
- **SQL interface**
- **ACID compliance**
- **Horizontal scalability – distributes database to many nodes**
- **Often uses Raft, Paxos consensus protocols**

Examples: Google Spanner, CockroachDB, YugabyteDB, TiDB

Time series databases

- **Columnar storage (most frequently)**
- **Offer time-based indexing**
- **Support very fast data inserts**
- **Efficiently compress the data**
- **Downsampling and retention policies to archive/remove old data, but keep aggregates**
- **Support for efficient window queries (e.g. last 10 minutes)**

Examples: InfluxDB, ClickHouse, TimescaleDB (TigerData)

Vector databases

“Two households, both alike in dignity”  *[0.329, 0.911, 0.21, 0.37, ...]*

The diagram shows a blue arrow pointing from the text "Two households, both alike in dignity" to a list of numbers [0.329, 0.911, 0.21, 0.37, ...]. The arrow is labeled "EMBEDDING MODEL" in white text.

Vector information is used for example for embeddings of documents in RAG systems

Most important operation – distance calculation, similarity search

Uses different types of indexes:

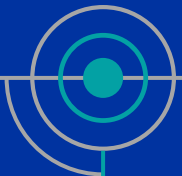
- **HNSW (hierarchical navigable small world)**
- **IVF (inverted file flat)**

Examples: Pinecone, Weaviate, Milvus, Chroma, Qdrant, Vespa.

But also:

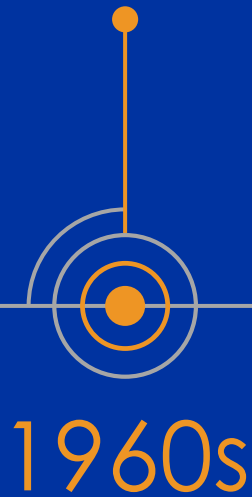
- **pgvector extension for Postgres**
- **VECTOR data type in Oracle, MySQL**

pre-1950s



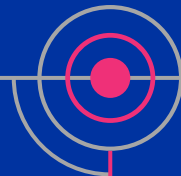
PHYSICAL RECORDS
PAPER IS KING!

FLAT FILES (E.G. CSV)



1960s

1970s



RELATIONAL
DATABASES

NOSQL DATABASES



2000s

2020s



NEWSQL & VECTOR
DATABASES